Formal verification of the index law and the resolvent lock in Lean 4

Supplement to "Odd-sector, Pontryagin-space and index-law extensions of the Connes–van Suijlekom real-zeros theorem" (version 2.2)

Supplement, version 1.0

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Abstract

Theorem 5(a) (index law, even sector) and Theorem 6 (resolvent lock) of [1] are formally verified in the Lean 4 proof assistant [2] with the Mathlib library [3]. For a real symmetric matrix Q of Connes–van Suijlekom type on $I = \{-N, \dots, N\}$, Theorem 5(a) states $N_c(\xi) + d(\xi) = \kappa(\lambda)$ for a simple eigenvalue λ with even eigenvector $\xi \notin \mathbb{R}e_0$, and Theorem 6 states $N_c(x) + d(x) + [x_0 < 0] = \kappa(\mu)$ for the resolvent vector $x = (Q - \mu)^{-1}e_0$: the zeros of a single polynomial count the eigenvalues below a level. Both theorems carry complete formal proofs; the axiom audit lists exactly the three standard axioms of Lean's foundations. The formal proofs follow the route of [1]—commutator identity, rank-one operator D'_η, Q-symmetry, norm identity and Pontryagin balance—and make three finite-dimensional steps explicit: the Sylvester bridge between eigenvalue counts and the negative index, the root count of a separable real polynomial, and the Pontryagin balance for a form whose kernel is spanned by one eigenvector. The library passes a negative control of 60 deliberately corrupted statements, continuous integration on an independent machine, and numerical cross-checks on 398 and 1210 random instances. The verified state is tag `v1.0`, commit `08a1c67`, of the public repository [4].

Scope. Formally verified (Lean 4 / Mathlib): Theorems 5(a) and 6 of [1], finite dimension. Measured: numerical cross-checks (Table 4). Continuation: Theorem 5(b) (odd sector) and the operator-level theorems of [1].

Keywords: formal verification, Lean 4, Mathlib, index law, resolvent lock, Pontryagin space, Connes–van Suijlekom class, real zeros, axiom audit.

1 Introduction

The note [1] proves, for real symmetric matrices of Connes–van Suijlekom (CvS) type [5], an index law (Theorem 5) and a resolvent lock (Theorem 6). Both read a spectral count from the zeros of a single polynomial; at $\mu = 0$ the resolvent lock is a zero-location criterion for positivity. This supplement records that Theorem 5(a) and Theorem 6 are formally verified in the Lean 4 proof assistant [2] with the Mathlib library [3]. The formal library is public [4]; the verified state is tag `v1.0`, commit `08a1c67`. Section 2 fixes the formal statements, Section 3 describes the architecture of the proof, Sections 4–5 report the verification protocol and the numerical cross-checks, and Section 6 lists the commands that reproduce every result.

2 Setting and statements

Let $N \geq 0$ and $I = \{-N, \dots, N\}$. A real matrix $Q = (q_{ij})_{i,j \in I}$ is of CvS type if $q_{ii} = a_i$ and $q_{ij} = (b_i - b_j)/(i - j)$ for $i \neq j$, with a even ($a_{-i} = a_i$) and b odd ($b_{-i} = -b_i$); such Q is

symmetric. Write $\beta = (b_i)_{i\in I}$, $\eta = (1,\dots,1)$, e_0 for the unit vector at 0, and for $\xi \in \mathbb{R}^I$

$$P_\xi(s) = \sum_{k\in I} \xi_k \prod_{j\in I,\, j\neq k} (j - s). \tag{1}$$

$N_c(\xi)$ is the number of pairs of non-real zeros of P_ξ, $d(\xi) = \#\{s \in \mathbb{R}\setminus\{0\} : P_\xi(s) = 0,\ P'_\xi(s)P_{\beta\odot\xi}(s) > 0\}$ is the defect of [1], and $\kappa(\mu)$ is the number of eigenvalues of Q below μ, counted with multiplicity. Table 1 lists the Lean counterparts.

Theorem 5(a) of [1] (`CvS.index_even`)**.** *Let Q be of CvS type and λ a simple eigenvalue with eigenvector ξ, where ξ is even and $\xi \notin \mathbb{R}e_0$, and let the zeros of P_ξ be simple and lie outside I. Then*

$$N_c(\xi) + d(\xi) = \kappa(\lambda). \tag{2}$$

Theorem 6 of [1] (`CvS.lock`)**.** *Let Q be of CvS type, let $\mu \in \mathbb{R}$ be distinct from all eigenvalues and $x = (Q-\mu)^{-1}e_0$, with $x_0 \neq 0$ and $\langle \eta, x\rangle \neq 0$, and let the zeros of P_x be simple and lie outside I. Then*

$$N_c(x) + d(x) + [x_0 < 0] = \kappa(\mu). \tag{3}$$

In the formal statements, simple zeros are expressed as separability of P_ξ (`Polynomial.Separable`), zeros outside I as $P_\xi(j) \neq 0$ for every $j \in I$, and simplicity of λ as $\dim\ker(Q-\lambda) = 1$ (`Module.finrank`).

Table 1: Notation of [1] and its counterpart in the formal library [4].

Notation in [1]	Lean counterpart (namespace `CvS`)
$I = \{-N,\dots,N\}$, label i	`Idx N` = `Fin (2*N+1)`, `lab`
CvS type: $q_{ii} = a_i$, $q_{ij} = \frac{b_i - b_j}{i-j}$	`IsCvS Q a b`
$P_\xi(s) = \sum_k \xi_k \prod_{j\neq k}(j-s)$	`P`
N_c	`Nc`
$d(\xi)$	`d`
$\kappa(\mu) = \#\{i : \lambda_i < \mu\}$	`Matrix.IsHermitian.eigenvalues`
$D'_\varphi = D - \lvert D\xi\rangle\langle\varphi\rvert/c$	`Dprime`
negative index of a form	`sigNeg`

3 Architecture of the formal proof

The formal proofs follow the route of [1]. Table 2 aligns each step with its place in [1] and with the Lean declarations that prove it. Three finite-dimensional facts that [1] uses as standard steps are formalized explicitly.

3.1 Sylvester bridge

For symmetric Q and every quadratic form q with $q(w) = w^\top(Q-\mu)w$, the negative index of q equals $\kappa(\mu)$ (`CvS.kappa_eq_sigNeg`). The proof diagonalizes Q in its orthonormal eigenbasis and applies Sylvester's law of inertia as it is available in Mathlib.

3.2 Degree and root count

The characteristic polynomial of D'_η satisfies $\chi_{D'_\eta} = c^{-1}X\,P_\xi$ with $c = \langle\eta,\xi\rangle$; hence $\deg P_\xi = 2N$ (`CvS.charpoly_lock`, `CvS.natDegree_P_lock`). For a separable real polynomial p,

$$\#\{\text{real zeros of } p\} + 2\,\#\{\text{zeros with } \operatorname{Im} > 0\} = \deg p \tag{4}$$

(`CvS.card_roots_split`), and N_c equals the number of zeros with positive imaginary part (`CvS.Nc_eq`).

3.3 Pontryagin balance with a one-dimensional kernel

In Theorem 5(a) the form $Q_\lambda = Q - \lambda$ is degenerate; [1] passes to the quotient $\mathbb{R}^I/\mathbb{R}\xi$, on which the induced form is non-degenerate. The formal proof uses the equivalent statement below.

Lemma F1 (`CvS.pontryagin_balance_ker`)**.** *Let B be a real symmetric $n \times n$ matrix and T a matrix with BT symmetric. Let $(r_i, v_i)_{i\in\iota}$ be real eigenpairs and $(\zeta_k, z_k)_{k\in K}$ non-real eigenpairs of T, with non-zero vectors, such that the values r_i, ζ_k, $\bar{\zeta}_k$ are pairwise distinct and $|\iota| + 2|K| = n$, and let $\ker B = \mathbb{R}v_{i_0}$ for one index i_0. Then the negative index of B equals $\#\{i : \ v_i^\top B v_i < 0\} + |K|$.*

In the application $v_{i_0} = \xi$ has norm 0 and contributes 0 to the count. Lemma F1 rests on Lemma 7 of [1] (the Pontryagin balance [6]) and on two facts proved with it: a vector B-orthogonal to a basis lies in $\ker B$ (`CvS.orth_all_ker`), and a B-orthogonal family whose norms are non-zero except at one non-zero member is a basis (`CvS.sigNeg_of_orth_family_ker`).

Table 2: Steps of the formal proof, their place in [1], and the Lean declarations that prove them.

Step	In [1]	Lean declaration(s)
$[D, Q] = \beta \wedge \eta$	Lemma 1	`commutator`, `commutator_shift`
$D'_\varphi \xi = 0$, $\det(D'_\varphi - s) = -\frac{s}{c} P_{\varphi\odot\xi}(s)$	Lemma 2	`Dprime_mulVec_self`, `det_Dprime`
Q_λ-symmetry criterion	Lemma 3	`Qlam_Dprime_isSymm`
Non-degeneracy, even sector	Lemma 4	`nondeg_even`
Norm identity	Lemma 6	`Qlam_norm`, `eig_sign`
Pontryagin balance	Lemma 7	`pontryagin_balance`
Eigenvalue count = negative index	proof of Lemma 7	`kappa_eq_sigNeg`
Degree and root count	proofs of Thms 5(a), 6	`charpoly_lock`, `natDegree_P_lock`, `card_roots_split`, `Nc_eq`
Balance with kernel $\mathbb{R}\xi$ (Lemma F1)	proof of Thm 5(a)	`pontryagin_balance_ker`, `orth_all_ker`, `sigNeg_of_orth_family_ker`, `simple_ker`
Lock setting: symmetry, eigenvectors, norms, signs	proof of Thm 6	`Qmu_Dprime_lock_isSymm`, `Dprime_eigvec_r`, `Dprime_eigvec_c`, `lock_norm_x`, `lock_norm`, `lock_sign`
Theorem 6 (resolvent lock)	Thm 6	`lock`
Theorem 5(a) (index law)	Thm 5(a)	`index_even`

4 Verification protocol

The library file `CvsIndexLaw/Lock.lean` has 2516 lines with 81 theorems and 10 definitions, built with Lean v4.36.0-rc1 and the Mathlib revision pinned in `lake-manifest.json`. Table 3 lists the checks. The axiom audit is decisive: `#print axioms` lists every axiom on which a theorem depends, and an open step appears as `sorryAx`. The audit file contains a deliberately open control statement, which the audit reports with `sorryAx`; the check therefore demonstrably detects open steps. The negative-control file holds 60 corrupted variants of the library's statements—flipped signs, dropped terms, wrong indices or coefficients—and Lean rejects each of them while accepting every correct line.

5 Numerical cross-checks

Independently of the formal proof, both identities were evaluated on random CvS matrices with $N = 1, \ldots, 5$ and standard normal a (even) and b (odd). For Theorem 6, μ is uniform in

Table 3: Verification protocol and results (tag `v1.0`).

Check	Command	Result
Build	`lake build`	completes; every declaration is closed by a complete proof
Axiom audit	`lake env lean kontrol/Aksiyom.lean`	`lock`, `index_even`: `propext`, `Classical.choice`, `Quot.sound`; control statement: `sorryAx`
Negative control	`lake env lean kontrol/NegatifKontrol.lean`	60 corrupted statements, 60 rejected
Independent build	GitHub Actions (Lean Action CI)	passed on `08a1c67` and `v1.0`
Repository content	`sorry` count in `Lock.lean`	0 at `08a1c67`; control: 1 at `34b2685`

$[\lambda_{\min} - 1, \lambda_{\max} + 1]$ at distance at least 10^{-3} from the spectrum; for Theorem 5(a), every simple eigenvalue with even eigenvector $\xi \notin \mathbb{R}e_0$ is used. Instances with a non-separable polynomial or a zero within 10^{-6} of I are set aside, as the hypotheses require. Table 4 reports the counts. The reversed sign condition ($P'P_{\beta\odot\xi} < 0$ in place of > 0) reproduces κ in 54 of 398 and in 166 of 1210 instances, so the identity separates the correct condition from its reverse. The resolvent lock has further been verified with interval arithmetic on truncated Weil quadratic forms and on an Epstein zeta function with genuine off-line zeros [1, 7].

Table 4: Numerical cross-check on random CvS matrices ($N = 1, \ldots, 5$; scripts in `kontrol/`). Reversed condition: $P'P_{\beta\odot\xi} < 0$.

Statement	Script	Instances	Identity holds	Reversed holds
Theorem 6	`kilit_teorem6_v1.py`	398	398	54
Theorem 5(a)	`indeks_teorem5a_v1.py`	1210	1210	166

6 Reproducibility

```
git clone https://github.com/akifoltu-ui/cvs-index-law-lean
cd cvs-index-law-lean
git checkout v1.0
lake exe cache get
lake build
lake env lean kontrol/NegatifKontrol.lean   # exactly 60 errors
lake env lean kontrol/Aksiyom.lean          # axiom audit
python3 kontrol/kilit_teorem6_v1.py
python3 kontrol/indeks_teorem5a_v1.py
```

7 Author's contributions

The statements and the proof route are those of [1]: the index law and the resolvent lock; the commutator identity $[D, Q] = \beta \wedge \eta$ of [5, Lemma 5.1] in the form used for the Q_λ-symmetry criterion (Lemmas 1 and 3 of [1]); the rank-one operator D'_φ with its determinant formula (Lemma 2); the non-degeneracy in the even sector (Lemma 4); the norm identity (Lemma 6); and the application of the Pontryagin balance (Lemma 7) to D'_η, on the quotient by $\mathbb{R}\xi$ for Theorem 5(a) and on all of $\mathbb{R}^I$ for Theorem 6. The formal library follows this route step by

step; Lemma F1 and Sections 3.1–3.2 make explicit the finite-dimensional facts that [1] uses. The verification protocol of Section 4 applies the author's standing rule that every check is accompanied by a deliberately corrupted input that the check must reject.

8 Priority

The formal verification of Theorems 5(a) and 6 of [1] was completed on 22 October 2026. The verified state is tag `v1.0`, commit `08a1c67`, of [4].

References

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